Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

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Team Size : 4

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Final Report

1. INTRODUCTION

1.1 Project Overview

This project focuses on analyzing housing market trends and factors affecting house prices using Tableau. By leveraging visualization techniques and data analysis, the project examines comprehensive housing data, including total sales by years since renovation, house age distribution by the number of bathrooms, bedrooms, and floors, and the impact of renovations on house age. The project aims to uncover insights that help stakeholders understand the housing market and make informed decisions

1.2 Purpose

The primary goal is to provide real-time, actionable insights into the housing market. By visualizing and interpreting patterns in house prices, sales trends, and property features, the project enables better pricing strategies, optimizes marketing decisions, and enhances overall market competitiveness. Key stakeholders include real estate analysts, marketing teams, and company executives

2. IDEATION PHASE

2.1 Problem Statement

Real estate stakeholders often face difficulty in understanding the factors driving house prices and sales trends. Without proper visualization and analysis, it is challenging to identify which features influence sales or how renovations and property characteristics impact pricing.

2.2 Empathy Map Canvas

- **Says:** "I want to know which houses sell better and why."

- **Thinks:** "If I understand sales trends, I can make smarter pricing and marketing decisions."

- **Does:** Collects housing data, tracks property sales, monitors renovations.
- **Feels:** Confused by the volume of data, unsure which features impact sales most.

2.3 Brainstorming

Use Tableau to visualize house sales trends and distributions
Analyze the impact of renovation and property features on prices
Create charts that make it easier for stakeholders to interpret housing market patterns

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- Collects housing data from records and databases
- Opens Tableau Dashboard
- Inputs filters like renovation years, number of bathrooms/bedrooms/floors
- Views house sales trends and distributions
- Makes strategic decisions based on visual insight

3.2 Solution Requirement

Input: **Housing dataset with features like age, renovation status, number of bedrooms, bathrooms, floors, and sales price**

Output: **Visualizations and insights for stakeholders to interpret housing trends**

Tool/Model: Tableau for data visualization

Platform: Desktop/Web Dashboard

Dataset Source: Public housing datasets, company records

3.3 Data Flow Diagram

Housing Dataset → Preprocessing/Transformation → Tableau Visualizations → Dashboard Display → Strategic Decision Making

3.4 Technology Stack

-Visualization Tool: Tableau
-Data Processing: Python (for preprocessing, optional)
-Database: CSV/SQL housing datasets
-Dataset Source: Kaggle housing datasets, company data

4. PROJECT DESIGN

4.1 Problem Solution Fit

Stakeholders need clear insights into housing trends and pricing factors
Tableau dashboard provides visual patterns and interactive analysis
Supports better decision-making in pricing and marketing

4.2 Proposed Solution

-Interactive Tableau dashboard for house sales trends and distributions
-Filters for renovation status, age, and features like bathrooms/bedrooms/floors
-Visualizations help identify patterns and optimize strategies

4.3 Solution Architecture

Housing Data Collection → Data Cleaning & Preprocessing → Tableau Dashboard → Interactive Visualizations → Stakeholder Analysis & Decision Making

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Week 1-2: Requirement gathering & dataset collection
Week 3-4: Data cleaning and preprocessing
Week 5-6: Tableau dashboard design and visualization creation
Week 7: Testing dashboard interactivity and accuracy
Week 8: Final deployment and documentation

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Tested usability and responsiveness of dashboard
Checked correct representation of total sales by years since renovation
Verified house age distribution and feature grouping
Ensured smooth filtering and interactive experience

7. RESULTS

7.1 Output Screenshots

-Dashboard Overview: Total sales and average prices
-Histogram: Total Sales by Years Since Renovation
-Pie Chart: House Age Distribution by Renovation Status
-Grouped Bar Chart: House Age Distribution by Bathrooms, Bedrooms, and Floors

8. ADVANTAGES & DISADVANTAGES

Advantages

-Quick insights into housing market trends
-Visualizes impact of renovations and house features on prices
-Interactive and user-friendly dashboard
-Helps stakeholders make data-driven decisions

Disadvantages

-Limited to features available in the dataset
-Accuracy depends on dataset quality
-Requires manual update when new data is added

9. CONCLUSION

Tableau visualizations effectively show housing market trends
Stakeholders can identify patterns in pricing, renovations, and house features
Supports strategic decision-making and planning

10. FUTURE SCOPE

Add more housing features like location, amenities, and neighborhood
Predictive analytics for house prices using ML models
Real-time dashboards with automated updates
Mobile integration for stakeholder access

11. APPENDIX

Dataset Link: https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2

GitHub & Project Demo Link : [https://github.com/gdivya1309/tableau]